THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · NCERT-TYPE CHAPTER TEST SET 2

Chapter 2: **Polynomials** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If α, β are zeroes of $x^2 - 5x - 24$, then $\alpha + \beta$ is: 1

(a) -5 **(b)** 6 **(c)** 5 **(d)** -24

2. If α, β are zeroes of $x^2 - 5x - 24$, then $\alpha\beta$ is: 1

(a) 24 **(b)** -23 **(c)** -24 **(d)** 5

3. When $x^2 - 5x + 4$ is divided by $x - 4$, the remainder is: 1

(a) 4 **(b)** 0 **(c)** 1 **(d)** 2

4. The sum of the zeroes of the cubic $x^3 - 7x^2 + 7x + 15$ is: 1

(a) 6 **(b)** 8 **(c)** 9 **(d)** 7

5. The number of zeroes of a polynomial whose graph cuts the x-axis at 3 points is: 1

(a) 2 **(b)** 4 **(c)** 5 **(d)** 3

6. If one zero of $x^2 - 3x - 4$ is -1, the other zero is: 1

(a) 5 **(b)** 4 **(c)** 6 **(d)** 3

7. A quadratic polynomial whose zeroes are 1 and 8 is: 1

(a) $x^2 + 9x - 8$ **(b)** $x^2 - 9x + 8$ **(c)** $x^2 + 9x + 8$ **(d)** $x^2 - 9x - 8$

8. If α, β are zeroes of $x^2 - 9x + 20$, then $\alpha + \beta$ is: 1

(a) 9 **(b)** -9 **(c)** 10 **(d)** 20

9. If α, β are zeroes of $x^2 - 9x + 20$, then $\alpha\beta$ is: 1

(a) 9 **(b)** -20 **(c)** 20 **(d)** 21

10. When $x^2 - 11x + 30$ is divided by $x - 5$, the remainder is: 1

(a) 1 **(b)** 5 **(c)** 0 **(d)** 6

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the sum and product of the zeroes of $3x^2 + 5x - 2$. 2

12. Find a quadratic polynomial with sum of zeroes 2 and product $\frac{1}{3}$. 2

13. The graph of $y = p(x)$ cuts the x-axis at $x = -3$ and $x = 1$ only. What are the zeroes of $p(x)$, and what is its likely degree? 2

14. Find the sum and product of the zeroes of $4x^2 - 4x - 3$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the zeroes of $x^2 + 3x - 10$ and verify the relationship between the zeroes and the coefficients. 3

16. Divide $x^3 - 2x^2 + 4x - 5$ by $x - 1$ and find the quotient and remainder. 3

17. Check whether $x + 1$ is a factor of $x^3 + x^2 + x + 1$ by division. 3

OR

Find the zeroes of $2x^2 + x - 6$ and verify the relationship between the zeroes and the coefficients. 3

18. Divide $3x^3 + 2x^2 - 7x + 1$ by $x + 2$ and find the quotient and remainder. 3

OR

Check whether $x - 2$ is a factor of $2x^3 - 5x^2 + x + 2$ by division. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Divide $x^4 + 2x^2 - 3$ by $x^2 + 1$ and verify the division algorithm. 5

OR

Given that 2 and -2 are two zeroes of $x^3 - 3x^2 - 4x + 12$, find the third zero. 5

20. Divide $x^4 + 3x^3 + 4x + 1$ by $x^2 + x + 1$ and verify the division algorithm. 5

OR

Given that 1 and -3 are two zeroes of $x^3 - 4x^2 - 15x + 18$, find the third zero. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND